Close

J B Dhejasvi-241801053-EX 1

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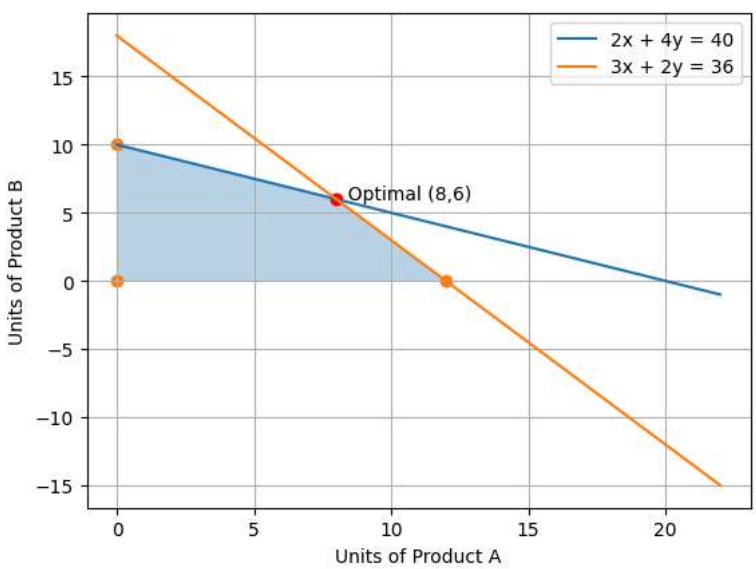
A factory manufactures two products A and B. Product A requires 2 hours of machine time and 3 kg of raw material per unit. Product B requires 4 hours of machine time and 2 kg of raw material per unit. Available machine time is 40 hours and raw material is 36 kg. Profit per unit is 50 for A and 70 for B. Formulate the LPP and solve it graphically.

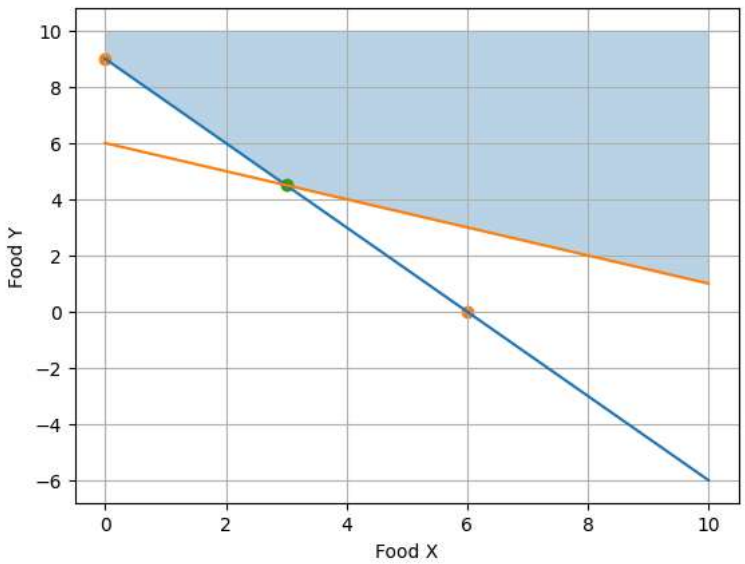
```
import numpy as np
import matplotlib.pyplot as plt


x = np.linspace(0, 22, 400)


y1 = (40 - 2*x) / 4      # 2x + 4y = 40
y2 = (36 - 3*x) / 2      # 3x + 2y = 36
plt.plot(x, y1, label='2x + 4y = 40')
plt.plot(x, y2, label='3x + 2y = 36')


plt.fill_between(x, np.minimum(y1, y2), where=(y1>=0) & (y2>=0), alpha=0.3)


points_x = [0, 12, 8, 0]
points_y = [0, 0, 6, 10]
plt.scatter(points_x, points_y)


plt.scatter(8, 6, color='red')
plt.text(8, 6, '  Optimal (8,6)', fontsize=10)

plt.xlabel('Units of Product A')
plt.ylabel('Units of Product B')
plt.legend()
plt.grid()
plt.show()
```



7. A person consumes two foods X and Y. Food X provides 3 units of protein and 1 unit of vitamin per unit. Food Y provides 2 units of protein and 2 units of vitamin per unit. Minimum requirements are 18 units of protein and 12 units of vitamins. Cost per unit is ■4 for X and ■3 for Y. Formulate and solve the LPP graphically

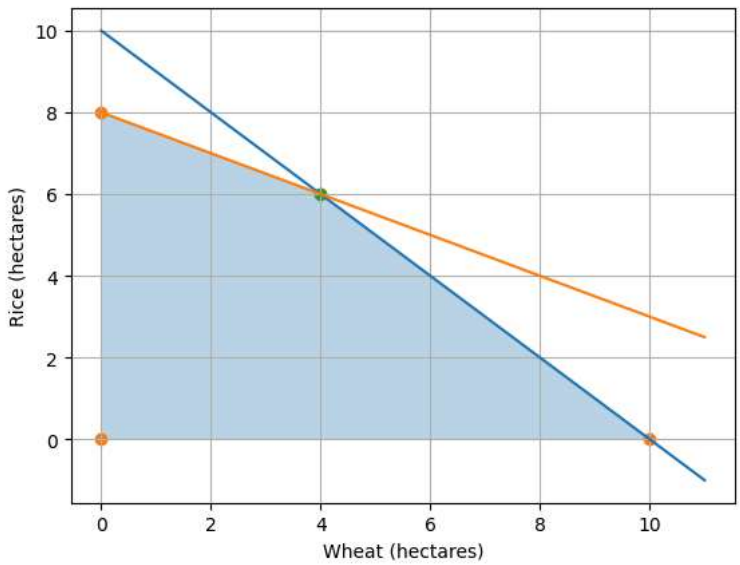
```
import numpy as np
import matplotlib.pyplot as plt
```

```
x = np.linspace(0, 10, 400)

y1 = (18 - 3*x) / 2
y2 = (12 - x) / 2

plt.plot(x, y1)
plt.plot(x, y2)

plt.fill_between(x, np.maximum(y1, y2), 10, where=(y1<=10) & (y2<=10), alpha=0.3)

plt.scatter([6, 0, 3], [0, 9, 4.5])
plt.scatter(3, 4.5)

plt.xlabel("Food X")
plt.ylabel("Food Y")
plt.grid()
plt.show()
```



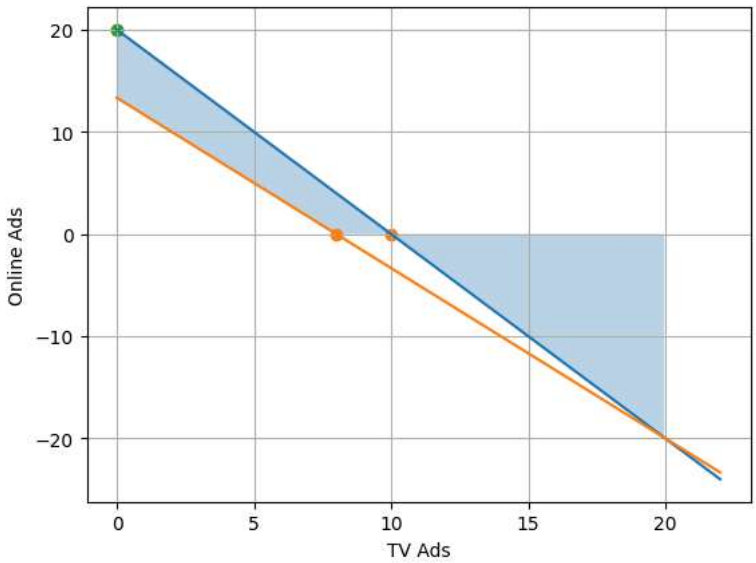
A farmer has 10 hectares of land to cultivate wheat and rice. Wheat requires 2 labour units per hectare and yields ■20,000 per hectare. Rice requires 4 labour units per hectare and yields ■30,000 per hectare. Total labour available is 32 units. Formulate and solve the LPP graphically

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 11, 400)

y1 = 10 - x
y2 = (32 - 2*x) / 4

plt.plot(x, y1)
plt.plot(x, y2)

plt.fill_between(x, np.minimum(y1, y2), where=(y1>=0) & (y2>=0), alpha=0.3)

plt.scatter([10, 0, 4, 0], [0, 8, 6, 0])
plt.scatter(4, 6)

plt.xlabel("Wheat (hectares)")
plt.ylabel("Rice (hectares)")
plt.grid()
plt.show()
```



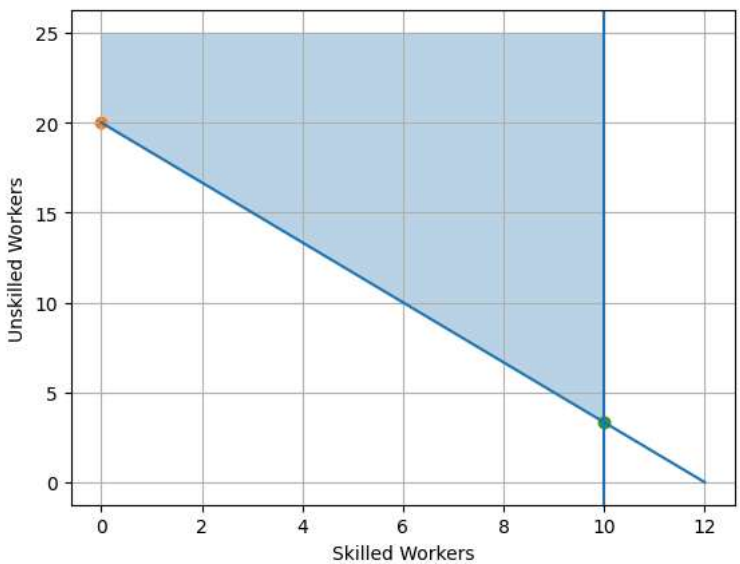
. A company advertises using TV and online media. A TV ad costs ■10,000 and reaches 50,000 people. An online ad costs ■5,000 and reaches 30,000 people. Advertising budget is ■1,00,000 and at least 4,00,000 people must be reached. Formulate and solve the LPP.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 22, 400)

y1 = 20 - 2*x
y2 = (40 - 5*x) / 3

plt.plot(x, y1)
plt.plot(x, y2)

plt.fill_between(x, np.maximum(y2, 0), y1, where=(y1>=y2), alpha=0.3)

plt.scatter([8, 10, 0], [0, 0, 20])
plt.scatter(0, 20)

plt.xlabel("TV Ads")
plt.ylabel("Online Ads")
plt.grid()
plt.show()
```



A company employs skilled and unskilled workers. A skilled worker produces 10 units/day at a cost of ■800/day, while an unskilled worker produces 6 units/day at a cost of ■500/day. Daily production must be at least 120 units and at most 10 skilled workers can be employed. Formulate and solve the LPP graphically

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 12, 400)

y1 = (120 - 10*x) / 6

plt.plot(x, y1)
plt.axvline(10)

plt.fill_between(x, y1, 25, where=(x<=10) & (y1<=25), alpha=0.3)

plt.scatter([0, 10], [20, 10/3])
plt.scatter(10, 10/3)

plt.xlabel("Skilled Workers")
plt.ylabel("Unskilled Workers")
plt.grid()
plt.show()
```



PART A (5 × 5 = 25 Marks)

1. Define a Linear Programming Problem and state its assumptions.
2. Explain the graphical method of solving an LPP and state its limitations.
3. Distinguish between feasible solution and optimal solution.
4. Write short notes on non-negativity constraints and bounded feasible region.
5. Explain the corner point principle in linear programming.

Answers for Part A:

A Linear Programming Problem (LPP) is a mathematical optimization problem in which a linear objective function is maximized or minimized subject to a set of linear constraints and non-negativity restrictions on decision variables. assumptions are:linearity, additivity, divisibility, certainty, non-negativity

2)The graphical method is used to solve LPPs involving two decision variables. The constraints are plotted as straight lines on a graph, and the feasible region is identified. The objective function is evaluated at the corner points of the feasible region to obtain the optimal solution.

--Steps

Convert inequalities into equations.

Plot each constraint on the graph.

Identify the feasible region.

Determine corner points.

Evaluate the objective function at each corner point.

Select the point giving the optimal value.

--Limitations

Applicable only to two variables.

Not suitable for complex or large problems.

Accuracy depends on graph scaling.

Cannot handle higher-dimensional problems.

3)Feasible Solution satisfies all constraints whereas Optimal Solution Gives best max/min value

4)Non-Negativity Constraints

These constraints require decision variables to be greater than or equal to zero

Bounded Feasible Region

A feasible region is bounded if it is enclosed within finite limits. In such cases, an optimal solution always exists at one of the corner points of the region.

5}The corner point principle states that if an LPP has an optimal solution, it will occur at one or more corner points (vertices) of the feasible region.